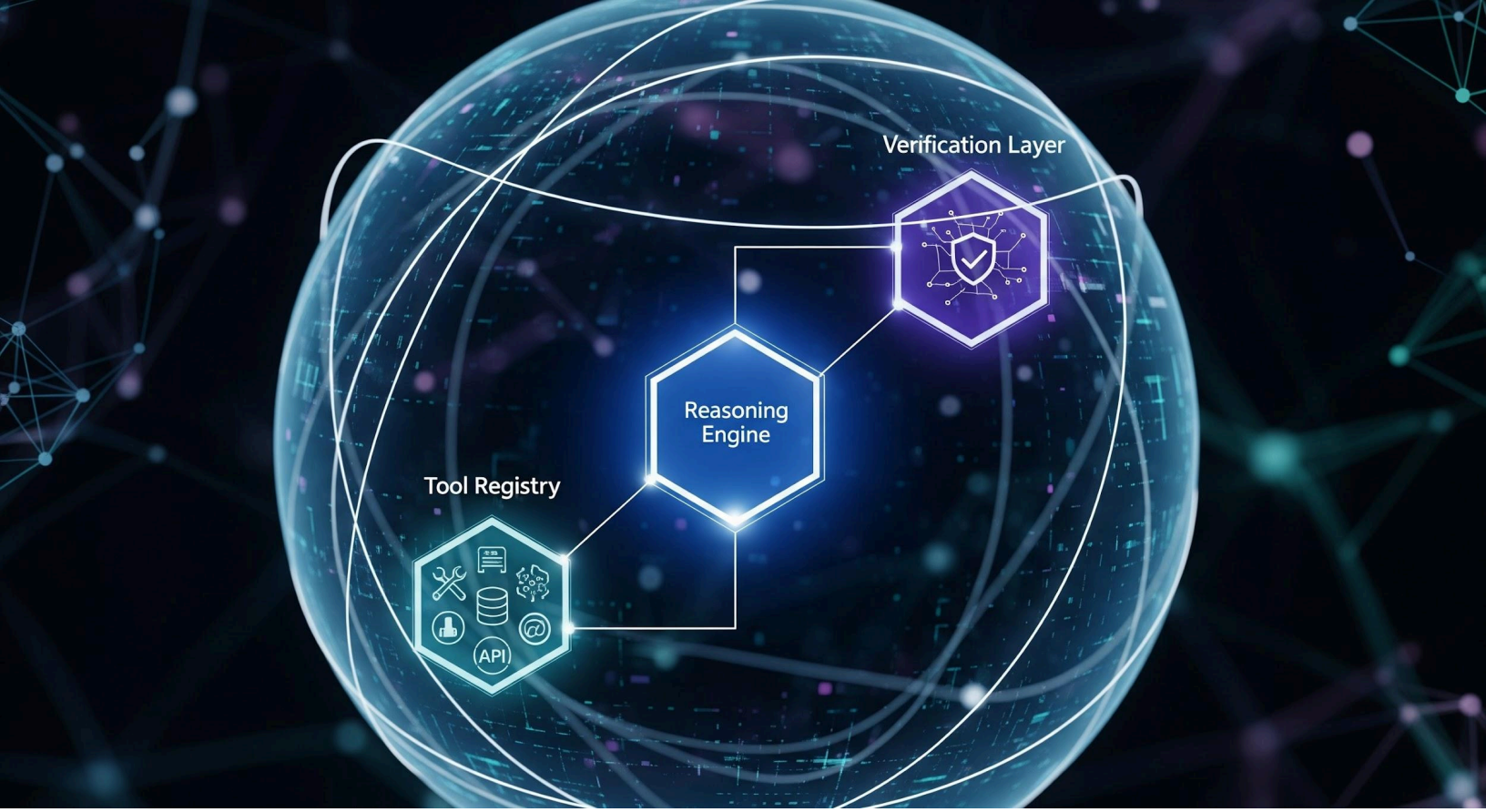

Pre-Search Checklist

Complete this before writing code. Save your AI conversation as a reference document.

Phase 1: Define Your Constraints

1. Domain Selection

- Which domain: healthcare, insurance, finance, legal, or custom?
- What specific use cases will you support?
- What are the verification requirements for this domain?
- What data sources will you need access to?

2. Scale & Performance

- Expected query volume?
- Acceptable latency for responses?
- Concurrent user requirements?
- Cost constraints for LLM calls?

3. Reliability Requirements

- What's the cost of a wrong answer in your domain?
- What verification is non-negotiable?
- Human-in-the-loop requirements?
- Audit/compliance needs?

4. Team & Skill Constraints

- Familiarity with agent frameworks?
- Experience with your chosen domain?
- Comfort with eval/testing frameworks?

Phase 2: Architecture Discovery

5. Agent Framework Selection

- LangChain vs LangGraph vs CrewAI vs custom?
- Single agent or multi-agent architecture?
- State management requirements?
- Tool integration complexity?

6. LLM Selection

- GPT-5 vs Claude vs open source?
- Function calling support requirements?
- Context window needs?
- Cost per query acceptable?

7. Tool Design

- What tools does your agent need?
- External API dependencies?
- Mock vs real data for development?
- Error handling per tool?

8. Observability Strategy

- LangSmith vs Braintrust vs other?
- What metrics matter most?
- Real-time monitoring needs?
- Cost tracking requirements?

9. Eval Approach

- How will you measure correctness?
- Ground truth data sources?
- Automated vs human evaluation?
- CI integration for eval runs?

10. Verification Design

- What claims must be verified?
- Fact-checking data sources?
- Confidence thresholds?
- Escalation triggers?

Phase 3: Post-Stack Refinement

11. Failure Mode Analysis

- What happens when tools fail?
- How to handle ambiguous queries?
- Rate limiting and fallback strategies?
- Graceful degradation approach?

12. Security Considerations

- Prompt injection prevention?
- Data leakage risks?
- API key management?
- Audit logging requirements?

13. Testing Strategy

- Unit tests for tools?
- Integration tests for agent flows?
- Adversarial testing approach?
- Regression testing setup?

14. Open Source Planning

- What will you release?
- Licensing considerations?
- Documentation requirements?
- Community engagement plan?

15. Deployment & Operations

- Hosting approach?
- CI/CD for agent updates?
- Monitoring and alerting?
- Rollback strategy?

16. Iteration Planning

- How will you collect user feedback?
- Eval-driven improvement cycle?
- Feature prioritization approach?
- Long-term maintenance plan?